Radionuclide Therapy for Bone Metastases

Utility of Scintigraphy and PET Imaging for Treatment Planning

Hojjat Ahmadzadehfar, MD, MSc[a,*], Markus Essler, MD[a], Kambiz Rahbar, MD[b], Ali Afshar-Oromieh, MD[c,d]

KEYWORDS

• Breast cancer • Prostate cancer • Lung cancer • Bone metastases • Radionuclide therapy • Radium-223 • PET

KEY POINTS

- A Pain relief response is seen in approximately one-half of patients treated with radionuclides for painful osseous metastases.
- FDG PET plays an important role in the detection of bony metastases with higher detection rate compared with bone scintigraphy in a majority of tumor types and consistent with clinical status of the patients as it reflects tumor activity.
- Skeletal tumor burden measured by fluoride PET correlates with OS inpatients who underwent bone targeted therapies using radionuclides.

TREATMENT OF BONE METASTASES WITH RADIONUCLIDES

The skeleton is a common site for metastases of various malignant tumors. Bone metastases are the result of complex interactions between tumor cells, bone cells, and the microenvironment.[1] Some tumor types, such as prostate cancer and breast cancer, are more frequently associated with bone metastases compared with others.[2] For inducing bone metastases, the tumor cells must detach from the primary tumor, enter the systemic circulation, evade detection by the immune system, and finally adhere to capillaries in the bone marrow, leading to extravasation into the bone marrow space.[1,3–5] Bone metastases are a major cause of morbidity and mortality and are associated with pain, pathologic fractures, spinal cord compression, and decreased survival.[6] The pain from small metastases seems to be caused by irritation of nerve endings in the endosteum by a variety of chemical mediators. Larger bone metastases produce stretching of the periosteum, which leads to pain.[7]

Bone pain palliation with radionuclides has a long history of using various β-emitters, like phosphorus-32 (^{32}P),[8] strontium-89 (^{89}Sr),[8] rhenium-186 hydroxyethylidene diphosphonate (^{186}Re-HEDP),[9] samarium-153 ethylenediamine tetramethylene phosphonate (^{153}Sm-EDTMP),[9] and, recently, lutetium-177 EDTMP (^{177}Lu-EDTMP)[10] and (4-{[(bis(phosphonomethyl))

[a] Department of Nuclear Medicine, University Hospital Bonn, Sigmund-Freud-Str. 25, Bonn 53127, Germany; [b] Department of Nuclear Medicine, University Hospital Muenster, Albert-Schweitzer-Campus 1, Muenster 48149, Germany; [c] Department of Nuclear Medicine, Heidelberg University Hospital, Im Neuenheimer Feld 400, Heidelberg 69120, Germany; [d] Clinical Cooperation Unit Nuclear Medicine, German Cancer Research Centre, Im Neuenheimer Feld 280, Heidelberg 69120, Germany

* Corresponding author. Department of Nuclear Medicine, University Hospital Bonn, Sigmund-Freud-Str. 25, Bonn 53127, Germany.

E-mail address: Hojjat.ahmadzadehfar@ukbonn.de

PET Clin 13 (2018) 491–503
https://doi.org/10.1016/j.cpet.2018.05.005
1556-8598/18/© 2018 Elsevier Inc. All rights reserved.

carbamoyl]methyl}-7,10-bis(carboxymethyl)-1,4,7,10-tetraazacyclododec-1-yl)acetic acid ^{177}Lu-BPAMD.[11] The only approved α-emitter is radium-223 (^{223}Ra).[12]

Bone-seeking radionuclides are classified into 2 groups: calcium analogs, like ^{89}Sr and ^{223}Ra, and radionuclides attached to phosphate, like ^{153}Sm-EDTMP and rhenium-186 hydroxyethylidene diphosphonate (^{188}Re-HEDP). The various radionuclides have different physical characteristics, which are shown in **Table 1**.[13] In this article, apart from introduction of different bone-specific radionuclide therapies, the value of PET imaging for the treatment planning and also its predictive value are discussed. Radionuclides that can be used for the treatment of all types of metastases (soft tissue, lymph nodes, and bone), such as ^{131}I for metastatic thyroid cancer,[14] ^{131}I-MIBG for neuroblastoma,[15] ^{177}Lu–prostate-specific membrane antigen (PSMA) for prostate cancer,[16] or ^{177}Lu-DOTATATE for neuroendocrine tumors.[17] These substances, however, are not addressed in this article.

BONE-TARGETED THERAPY WITH BONE SPECIFIC β-EMITTERS

Phosphorus-32 and strontium-89

Typical response times to different β-emitters are shown in **Box 1** and **Table 2**. Pain reduction after the injection of ^{32}P and ^{89}Sr occurs after 5 days to 14 days and 14 days to 28 days postinjection, respectively. During treatment with ^{32}P, pain relief was reported by 50% to 87% of prostate cancer patients treated with 200 MBq to 800 MBq of ^{32}P administered in daily fractions of 20 weeks to 80 MBq after androgen priming. The main disadvantage of ^{32}P therapy is a dose-limiting reversible pancytopenia due to myelosuppression with a maximum at 5 weeks to 6 weeks after administration.[18]

The first studies using ^{89}Sr have demonstrated an efficacy for pain reduction as high as 80%.[19–24] A majority of treated patients in these studies suffered from prostate or breast cancer. ^{89}Sr was the first radiopharmaceutical to be approved for systemic radionuclide therapy in the palliation of painful bone metastases. In the largest study using ^{89}Sr, 622 patients were included. Of these, 15% showed complete pain relief and a partial response was documented in 81%.[25,26] In only 4% of the patients, no response was observed. In a randomized, controlled phase III trial, James and colleagues[27,28] compared the efficacy of zoledronic acid (ZA) with ^{89}Sr regarding clinical progression-free survival (CPFS) as the primary outcome and the skeletal-related event–free interval (SREFI), total skeletal-related events (SREs), overall survival (OS), and quality of life (QOL) as secondary outcomes. Each of the compounds (ZA and ^{89}Sr) was combined with docetaxel for the treatment of bone metastases of castration-resistant prostate cancer (CRPC) patients. In this trial, 757 patients were randomized to receive 6 cycles of docetaxel combined with prednisolone, with ZA, with a single ^{89}Sr dose after the sixth cycle of docetaxel, or with both ZA and a single ^{89}Sr dose. A Cox regression analysis adjusted for stratification variables and showed a CPFS benefit for ^{89}Sr (hazard ratio [HR] = 0.845, P = .036) and confirmed no effect of ZA (P = .46). ZA showed a significant SREFI effect (HR = 0.76; P = .008), however. Neither agent affected OS (^{89}Sr, P = .74; ZA, P = .91). QOL was well maintained in all treatment arms, with differing patterns of care resulting from the effects of ^{89}Sr on the time to progression and ZA on SREFI and total SREs. The investigators concluded that ^{89}Sr combined with docetaxel improved CPFS but did not improve OS, SREFI, or total SREs. Furthermore, ZA did not improve CPFS or OS

Table 1
Summary of main physical properties of different radionuclides clinically used for pain palliation

	Emission Type	Half-Life (d)	Maximum Energy (MeV)	Maximum Tissue Penetration Range (mm)
^{32}P	ß-emitter	14.3	1.7	8.5
^{89}Sr	ß-emitter	50.5	1.46	7
Samarium-153	ß and γ-emitter	1.9	0.81	4
^{186}Re	ß and γ-emitter	3.7	1.07	5
^{188}Re	ß and γ-emitter	0.7	2.1	10
^{177}Lu	ß and γ-emitter	6.7	0.498	1.8
^{223}Ra	α-Emitter and γ-emitter	11.4	27.78	0.1

Adapted from Ahmadzadehfar H. Targeted therapy for metastatic prostate cancer with radionuclides. In: Mohan R, editor. Prostate cancer - leading-edge diagnostic procedures and treatments. Croatia: InTech; 2016.

Box 1
Take-home points

Several ß-emitters and 1 α-emitter are currently available for treatment of metastatic bone pain. They all show low toxicity and high therapeutic effects.

Osteoblastic metastases show a significantly better response to bone-targeted therapies compared with osteolytic metastases.

A comparison between the α-emitter and ß-emitters has not yet been conducted.

Pain reduction can be expected in 80% of the patients after the first treatment.

Pain reduction can be expected within several days following radionuclide therapy and can last for several months.

Interval treatments showed to be more effective compared with 1 single injection.

Treatment planning is regularly conducted with bone scintigraphy (lower costs compared with PET).

The BSI was shown associated with OS after radionuclide bone-targeted therapy.

Quantification tools in fluoride PET can help to predict the OS of patients treated with radionuclides.

In selected tumor entities, FDG PET demonstrated superiority compared with other methods for diagnosis of bone metastases.

For treatment planning and response assessment of bone metastases of prostate cancer, PSMA PET seems to be a promising method compared with other modalities.

but significantly improved the median SREFI and reduced the total SREs by approximately one-third, suggesting a role in postchemotherapy maintenance therapy.

Several patient characteristics could predict a favorable response to ^{89}Sr. A normal serum hemoglobin (Hb) level prior to treatment is associated with a higher pain response rate.[29] Predictors of a poor pain response include a low performance status, a higher serum prostate-specific antigen (PSA), more extensive osseous metastases, and a poor PSA response.[30–33] The toxicity of ^{89}Sr is limited, with the common development of thrombocytopenia, with a nadir between the fourth week and sixth week. Recovery of thrombocytopenia is typically slow over the next 6 weeks and is dictated by the skeletal tumor extent and bone marrow reserve.[34]

Rhenium-186 hydroxyethylidene diphosphonate and Rhenium-188 hydroxyethylidene diphosphonate

Pain reduction after the injection of ^{186}Re-HEDP occurs 2 days to 7 days after application, and this effect lasts approximately 8 weeks to 10 weeks. Previous studies have shown a pain relief response rate of 60% to 86%.[31,35–37] Toxicity has been limited to temporary myelosuppression, with platelet and neutrophil nadirs at 4 weeks after therapy. Recovery occurs within 8 weeks and is usually complete.[38] Pain reduction after the injection of ^{188}Re-HEDP occurs 2 days to 7 days after application and this effect lasts approximately 8 weeks.[13] The overall response rate is 69% to 77%.[39–41] ^{188}Re is of special interest in clinical applications because of its excellent availability and cost-effectiveness, because it is the product of a ^{188}W (^{188}W/^{188}Re) generator.[41] The short physical half-life (0.7 days) and high dose rate are predicted to lead to a rapid symptom response. Fractionated therapy has been shown to prolong the response duration and progression-free survival (PFS).[42]

Table 2
Bone-seeking radionuclides

Radiopharmaceutical	Activity	Typical Response Time	Typical Response Duration	Retreatment Interval
^{32}P	444 MBq (fractionated)	14 d	10 wk	>3 mo
^{89}Sr-chloride	150–200 MBq	14–28 d	12–26 wk	>3 mo
^{153}Sm -EDTMP	37 MBq/Kg	2–7 d	8 wk	>2 mo
^{186}Re-HEDP	1.3 GBq	2–7 d	8–10 wk	>2 mo
^{188}Re-HEDP	1.3–4.4 GBq	2–7 d	8 wk	2 mo
^{223}Ra	55 KBq/Kg body weight	1–8 wk	6 wk	6 injections every 4 wk

Data from Refs.[34,104,105]

Palmedo and colleagues[43] showed that 2 injections of ^{188}Re-HEDP with an 8-week interval were more effective for pain palliation compared with only one injection, with response rate and time of response of 92% and 5.7 months, respectively (P = .006 and P = .001, respectively). Biersack and colleagues[42] also reported a positive effect of repeated therapy using ^{188}Re-HEDP on OS. They retrospectively analyzed 60 CRPC patients who had been assigned to 3 different groups according to the number of therapies. The mean survival after the initial therapy improved from 4.5 months in patients with only 1 therapy to 9.98 months in patients with 2 therapies, and 15.7 months in patients with 3 or more cycles of therapy.[42] A randomized phase II study of repeated ^{188}Re-HEDP cycles combined with docetaxel/prednisolone versus docetaxel and prednisone alone in CRPC patients has shown that the combined treatment with ^{188}Re-HEDP and docetaxel did not prolong PFS. The median OS was significantly longer in the ^{188}Re-HEDP group (33.8 months), than in the control group (docetaxel/prednisolone: 21.0 months; P = .012).[44] This OS prolongation warrants further studies in the combined treatment of chemotherapy and radiopharmaceuticals. Moderate thrombopenia and leukopenia have been reported in different studies.[41,43–45]

Samarium-153 ethylenediamine tetramethylene phosphonate

^{153}Sm-EDTMP is one of the most widely used radionuclides for bone palliation. Since the clinical introduction of ^{223}Ra, however, it has been underestimated and not used as often as it could be. Pain reduction after the injection of ^{153}Sm-EDTMP occurs 2 days to 10 days after the application. Previous studies have shown a pain relief response rate of 59% to 86%.[46–50] Larger studies with more than 100 patients have demonstrated a median therapeutic efficacy of 80%. In a randomized, double-blind, placebo trial (n = 152), pain relief occurred in 65% of patients after ^{153}Sm-EDTMP treatment compared with 45% in the placebo group.[48] A significant decrease in pain between the ^{153}Sm-EDTMP and placebo groups was reported after 1 week and the analgesic intake was significantly reduced after 3 weeks and 4 weeks. ^{153}Sm-EDTMP can be administered prior to chemotherapy with docetaxel in CRPC patients. It neither reduces the antitumor efficacy of docetaxel nor causes additional toxicities.[51] Various studies with different protocols have shown the feasibility of a combination therapy of ^{153}Sm-EDTMP with the chemotherapeutic agent docetaxel in CRPC patients.[52–54] These studies have concluded that ^{153}Sm-EDTMP can be safely combined with docetaxel in patients with CRPC. The authors, however, do not currently recommend this approach because of a new generation of antiandrogenic therapeutics, such as enzalutamide or abiraterone, which may be combined with β-emitters with a lower probability of hematotoxicities.[55]

Lutetium-177 ethylenediamine tetramethylene phosphonate

There is increasing evidence for the utility of ^{177}Lu labeled with different compounds for the treatment of various tumors, like neuroendocrine tumors and prostate cancer.[56–58] This is mainly due to the favorable physical characteristic of ^{177}Lu, including a short tissue penetration range, which reduces the probability of bone marrow suppression while simultaneously delivering an appropriate dose to the target due to its long half-life of 6.73 days.[59]

Alavi and colleagues[60] reported that, in 30 patients treated with ^{177}Lu-EDTMP, the pain relief began 2.53 ± 2.08 weeks after the injection, and the response lasted for 4.38 ± 3.34 weeks. Among these patients, 16 (53%) showed a complete palliative pain response, 9 (30%) revealed a partial response, and 5 (17%) showed no response to treatment. Thus, a total treatment response was achieved in 25 patients (83%).

In a phase II study, Yuan and colleagues[10] treated 16 breast cancer and prostate cancer patients with ^{177}Lu-EDTMP. The patients were divided into 2 groups according to the administered activity (group 1 with 1295 MBq vs group 2 with 2590 MBq). The rate of complete responses in bone pain palliation was 55% in group 1% and 80% in group 2 at 6 weeks after treatment. In group 1, 1 patient experienced a grade III toxicity in both the Hb and platelet counts; however, no grade IV toxicities were observed. In group 2, there were no grade III or IV toxicities in Hb, platelet, or leukocyte counts. Moreover, no clinically significant adverse effects were observed, and no significant differences in either efficacy or safety were detected between the 2 dose levels.[10] Another phase II study by Agarwal and colleagues[61] included 44 patients with CRPC and breast cancer treated with ^{177}Lu-EDTMP at a dose of 1295 MBq (group 1) and 2590 MBq (group 2). The overall response rate (in all 44 patients) was 86%, in group 1 the response rate was 77%, and it was 95% in group 2 (P = .188). Complete, partial, and minimal responses were seen in 6 patients (13%), 21 patients (48%), and 11 patients (25%), respectively. A favorable response was seen in 27 patients

(84%) with prostate cancer and in 11 patients (92%) with breast cancer. There was no statistically significant difference in response rates between the prostate and breast cancer patients ($P = .9$). No serious hematological toxicity (grade I/II) was observed in 15 patients (34%), whereas a serious toxicity (grade III/IV) occurred in 10 patients (23%). There was no statistically significant difference in hematological toxicity between the groups.

COMPARING THE PAIN RESPONSE BETWEEN DIFFERENT β-EMITTERS

Dickie and Macfarlane[62] compared ^{89}Sr with ^{153}Sm in 57 prostate cancer patients. They found no difference in the pain response rate and toxicity. Baczyk and colleagues[63] did a similar comparison between these radionuclides in 100 prostate and breast cancer patients. They reported that the analgesic effects of ^{89}Sr and ^{153}Sm-EDTMP were similar in both prostate and breast carcinoma patients.

van der Poel and colleagues[24] compared ^{186}Re-HEDP with ^{89}Sr and reported no differences in the response rate or toxicity. A nonrandomized comparison of ^{188}Re-HEDP and ^{153}Sm-EDTMP in patients with painful metastases from prostate and breast cancer by Liepe and colleagues[64] showed a comparable response and toxicity with both agents. Liepe and Kotzerke[9] also performed a comparative study of ^{188}Re-HEDP, ^{186}Re-HEDP, ^{153}Sm-EDTMP, and ^{89}Sr in the treatment of painful bone metastases. They reported that all radiopharmaceuticals were effective in pain palliation, without the induction of severe side effects or significant differences in therapeutic efficacy or toxicity. In a prospective study, Sharma and colleagues[65] compared the therapeutic efficacy of ^{153}Sm-EDTMP and ^{177}Lu-EDTMP in 30 patients suffering from breast and prostate cancer. Both ^{153}Sm-EDTMP and ^{177}Lu-EDTMP delivered similar absorbed doses to the metastatic sites. The mean absorbed doses to various other organs were comparable and within the safe limits. A complete response for each radionuclide was evaluated as 80.0%. No significant alteration in blood parameters and no untoward reactions were observed. In another study, Thapa and colleagues[66] showed a similar response efficacy between ^{153}Sm-EDTMP and ^{177}Lu-EDTMP.

BONE-TARGETED THERAPY WITH THE BONE-SPECIFIC α-EMITTER RADIUM-223 DICHLORIDE

The first, and currently the only, Food and Drug Administration–approved radiopharmaceutical for the treatment of symptomatic bone metastases of CRPC patients with a positive impact on OS is ^{223}Ra-dichloride.[67] It has a half-life of 11.4 days, with a total emitted energy of approximately 28 MeV. Radium-223 (^{223}Ra) has a very high linear energy transfer and a very short penetration range in tissue (<100 μm). Therefore, despite the high energy, because of the short penetration range, bone marrow damage is minimal.[68] Double-strand breaks are induced even in quiescent cells and at low oxygen levels.[69] Thus, the α-particle is considerably more destructive to tumor cells than the β-particle.

In the randomized phase III Alpharadin in Symptomatic Prostate Cancer (ALSYMPCA) trial, 921 patients with progressive and symptomatic bone metastases with at least 2 or more metastases on bone scintigraphy and no known visceral metastases were recruited. Randomization was 2:1 in a double-blind fashion to receive 6 cycles of intravenous ^{223}Ra every 4 weeks with the best standard of care, or 6 infusions of placebo with the best standard of care. After treatment, there was a significant prolongation of OS (14.9 vs 11.3 months; $P<.001$), the frequency of SREs was reduced in the ^{223}Ra group, and the median time to a SRE increased (15.6 vs 9.8 months; $P<.001$). ^{223}Ra patients started subsequent chemotherapy later than placebo patients, and the chemotherapy duration was similar between the groups. In the ^{223}Ra and placebo patients receiving subsequent chemotherapy, the median hematological values (Hb, neutrophils, and platelets) remained nearly constant for up to 18 months after the start of chemotherapy, regardless of prior docetaxel treatment.[70] ^{223}Ra was well tolerated with low rates of grade 3/4 neutropenia (1.8% vs 0.8%) and thrombocytopenia (4% vs 2%).[12] An improvement of OS with ^{223}Ra was accompanied by significant QOL benefits, including a higher percentage of patients with a meaningful QOL improvement and a slower decline in QOL over time in patients with CRPC.[71]

In an open-label, phase IIIb trial,[55] 839 patients with CRPC were enrolled and, of these, 696 received at least 1 dose of ^{223}Ra, and 403 (58%) of these patients received the targeted 6 injections. The median follow-up period was 7.5 months. The investigators reported a median OS of 16 months. The median OS was longer in patients: with baseline alkaline phosphatase (ALP) concentration less than the upper limit of normal than for patients with ALP concentration equal to or greater than the upper limit of normal (median not reached vs 12 months); patients with baseline Hb levels of greater than or equal to 10 g/dL than for patients with Hb levels less than 10 g/dL (median 17 months vs 10 months); patients with a baseline Eastern

Cooperative Oncology Group (ECOG) performance status (PS) of 0 (median: not reached) than for patients with an ECOG PS of 1 (median: 13 months) or an ECOG PS of 2 or more (median: 7 months); and for patients with no reported baseline pain (median: not reached) than for those with mild pain (median: 14 months) or moderate to severe pain (median: 11 months, range: 9–13). The median OS was longer in patients who received ^{223}Ra plus abiraterone or enzalutamide, or both (median not reached), compared with those who did not receive these therapies (median: 13 months). The median OS was also longer in patients who received ^{223}Ra plus denosumab (median: not reached) compared with patients who received ^{223}Ra without denosumab (median: 13 months). Furthermore, the efficacy of ^{223}Ra in patients who were asymptomatic at baseline was well demonstrated by the investigators. The most common grade 3 or worse treatment-related adverse events were anemia in 5%, thrombocytopenia in 2%, neutropenia in 1%, and leukopenia in 1% of the patients. Any grade of serious adverse events was reported in 35% of the patients. The most common side effects, some of which were dose related, were diarrhea, fatigue, nausea, vomiting, and bone pain.[72–74] These side effects are easily manageable with symptomatic and supportive treatments.[75] Sartor and colleagues[76] have shown the feasibility of retreatment with ^{223}Ra in 44 patients who had undergone 6 initial ^{223}Ra injections. In this study, 66% of the patients received all 6 retreatment injections. No grade 4/5 hematologic treatment emergent adverse events occurred. The median OS and time to the first skeletal event were not reached.

Etchebehere and colleagues[77] retrospectively reviewed 110 patients with metastatic prostate cancer treated with ^{223}Ra. The end points of the study were OS, bone event–free survival (BeFS), PFS, and bone marrow failure. They evaluated the following parameters prior to the first therapy cycle: Hb, PSA, ALP, ECOG status, pain score, prior chemotherapy, and external beam radiation therapy (EBRT). Furthermore, during or after ^{223}Ra, the PSA doubling time (PSADT), the total number of radium cycles (RaTot), the use of chemotherapy, EBRT, enzalutamide, and abiraterone were evaluated. A significant reduction of ALP and pain scores occurred throughout the ^{223}Ra cycles. The risk of progression was associated with a declining ECOG status and a decrease in PSADT. RaTot, the initial ECOG status, ALP, the initial pain score, and the use of abiraterone were associated with OS ($P \leq .008$), PFS ($P \leq .003$), and BeFS ($P \leq .020$). RaTot, the initial ECOG status, ALP, and the initial pain score were significantly associated with bone marrow failure ($P \leq .001$), as well as Hb (<.001) and EBRT ($P = .009$). In the multivariate analyses, only RaTot and abiraterone remained significantly associated with OS ($P < .001$ and $P = .033$, respectively), PFS ($P < .001$ and $P = .041$, respectively), and BeFS ($P < .001$ and $P = .019$, respectively). Additionally, RaTot ($P = .027$) and EBRT ($P = .013$) remained significantly associated with bone marrow failure. They concluded that the concomitant use of abiraterone and ^{223}Ra seems to have a beneficial effect, while the EBRT may increase the risk of bone marrow failure.

THE PROGNOSTIC UTILITY OF RADIONUCLIDE IMAGING FOR BONE-TARGETED THERAPY

As discussed previously, osteoblastic metastases show a better response to the bone-targeted therapies compared with osteolytic metastases. Different studies have demonstrated the prognostic value of quantitative assessments of bone scintigraphy and ^{18}F-PET for prognostication of clinical outcomes and hematological toxicity in patients treated with bone-specific radionuclides.

Prognostic Value of Bone Scintigraphy

Quantification of skeletal tumor burden on bone scintigraphy can be done by using the EXINI bone software (EXINI Diagnostics, Lund, Sweden), which calculates the percentage of total skeletal mass affected by metastases as the bone scan index (BSI). Fosbol and colleagues[78] reported the utility of the BSI as a prognostic biomarker of OS and hematological toxicity as well as a tool for the response assessment in 88 patients with CRPC treated with ^{223}Ra. BSI was significantly associated with OS in multivariate analysis. The median OS for patients with a BSI greater than 5 was 8.2 months versus 15 months for patients with a BSI less than or equal to 5 ($P = .001$). The baseline BSI was prognostic for the occurrence of hematological toxicity, and patients with a BSI greater than 5 had an odds ratio of 3.02 ($P = .02$) for toxicity. The BSI declined during therapy in 44% of patients who completed 3 cycles of ^{223}Ra ($n = 52$), and in 84% of patients after the end of therapy (n = 32). There was no significant association between BSI changes during therapy and OS. They concluded that the BSI is a promising biomarker for late-stage metastatic CRPC patients receiving ^{223}Ra for prognostication of OS and hematological toxicity.

Prognostic Value of PET

A quantification of a fluoride PET is easier than bone scintigraphy. Rohren and colleagues[79]

described a quantification method to assess the skeletal tumor burden with fluoride PET scans. They used the volume of interest (VOI) on whole-body scans and adjusted the maximum standardized uptake value (SUV_{max}) at 10 (according to the established threshold of normal bone uptake), which excludes 99% of all normal bone uptake. They determined the skeletal tumor burden by calculating the fluoride tumor volume within the VOI (FTV_{10}) and the total lesion fluoride uptake as a product of the mean $SUV_{max} \times$ VOI (TLF_{10}).[79] The investigators evaluated the role of the skeletal tumor burden on baseline fluoride PET in predicting OS in patients who had undergone therapy with ^{223}Ra.[80] In this study, Etchebehere and colleagues[80] defined TLF_{10} and FTV_{10} in 42 CRPC patients. TLF_{10} and FTV_{10} at baseline were highly correlated and significant independent predictors of OS ($P = .0212$; HR = 5.990). A TLF_{10} cutoff value of 8000 discriminated the survivors from nonsurvivors after ^{223}Ra (with TLF_{10} values < 8,000, the median OS was not reached, whereas with TLF_{10} > 8,000, the median OS was 6.67 months).

^{18}F-fludeoxyglucose (FDG) PET/CT recognizes bone and bone marrow metastases with a high sensitivity in a majority of cancers compared with other modalities.[81–85]

It has been reported that lytic lesions have a higher SUV and are better detected by FDG PET than sclerotic lesions in patients with progressive disease.[86] Most lytic bone lesions heal with sclerosis, and FDG activity is expected to decrease or resolve after effective treatment.[84,87] At the same time these osteosclerotic lesions could be positive on bone scan suggesting that bone remodeling may continue for an extended period. Therefore, FDG PET may be a better indicator of active bony metastases.[84]

Compared with FDG PET/CT, FDG PET/MR imaging may offer a higher reader confidence without a significant difference in overall detection rate of bony metastases.[88]

In a meta-analysis of 17 article, including 2940 patients suffered from lung cancer, Qu and colleagues[85] reported that both ^{18}FDG PET/CT and ^{18}FDG PET were better imaging methods for diagnosing bone metastasis from lung cancer than MR imaging and bone scintigraphy. ^{18}FDG PET/CT has higher diagnostic value (sensitivity, specificity, and diagnostic odds ratios) for diagnosing bone metastasis from lung cancer than any other imaging methods.

The pooled sensitivity for the detection of bone metastasis in lung cancer using ^{18}FDG PET/CT, ^{18}FDG PET, MR imaging, and bone scintigraphy were 92%, 87%, 77% and 86%, respectively. The pooled specificity using ^{18}FDG PET/CT, ^{18}FDG PET, MR imaging, and bone scintigraphy were 98%, 94%, 92% and 88%, respectively.

In breast cancer, different meta-analysis showed the superiority of FDG PET/CT compared with bone scintigraphy in detection of bone and bone marrow metastases, with a sensitivity of 81% to 90% and specificity of more than 90%.[89–91] Regarding treatment response assessment of bone metastases in breast cancer, it has been reported that FDG PET/CT compared with bone scintigraphy is a powerful tool and consistent with clinical status of the patients because it reflects tumor activity. The investigators showed that bone scintigraphy is insufficient for response assessment of bone metastases because it reflects osteoblastic reaction of the bone against metastatic disease, which increases as the disease responds to treatment.[92]

In prostate cancer, most skeletal metastases have a low glycolytic rate, especially in the hormone-sensitive status of the disease. This could be one of the reasons for controversial reports of different studies regarding the detection rate of FDG PET in prostate cancer bony metastases. Morris and colleagues[93] reported that FDG PET can discriminate active osseous disease from scintigraphically quiescent lesions in patients with progressive metastatic prostate cancer. They showed that post-treatment changes in the SUV tend to correlate with changes in PSA.

Two other studies, however, reported lower sensitivity of FDG PET compared with bone scintigraphy in the detection of bony metastases[94,95]

PET/CT using ^{68}Ga-labelled ligands targeting the PSMA is a sensitive and specific diagnostic modality in prostate cancer diagnostic imaging.[96–98] Bräuer and colleagues[99] reported that ^{68}Ga-PSMA PET/CT revealed, in 15% of patients evaluated for a therapy with ^{223}Ra, previously unknown visceral metastases. In addition, ^{68}Ga-PSMA PET/CT revealed more extended tumor involvement in the bone compared with bone scintigraphy in 9 patients (33%). One other study[100] has demonstrated the utility of ^{68}Ga-PSMA PET as a gatekeeper for the treatment planning of metastatic prostate cancer with ^{223}Ra. The investigators showed that ^{68}Ga-PSMA PET can better select eligible patients for a bone-targeted therapy compared with a bone scan 63 patients who had undergone a total of 307 cycles of therapy with ^{223}Ra were analyzed. In 31 patients, bone scans and radiological imaging were performed for pretherapeutic imaging (group 1). In 32 patients, bone scans and ^{68}Ga-PSMA PET were performed before therapy (group 2). Patients with small lymph node metastases as well as local recurrence were not excluded from treatment, consistent with current guidelines. PSA and ALP

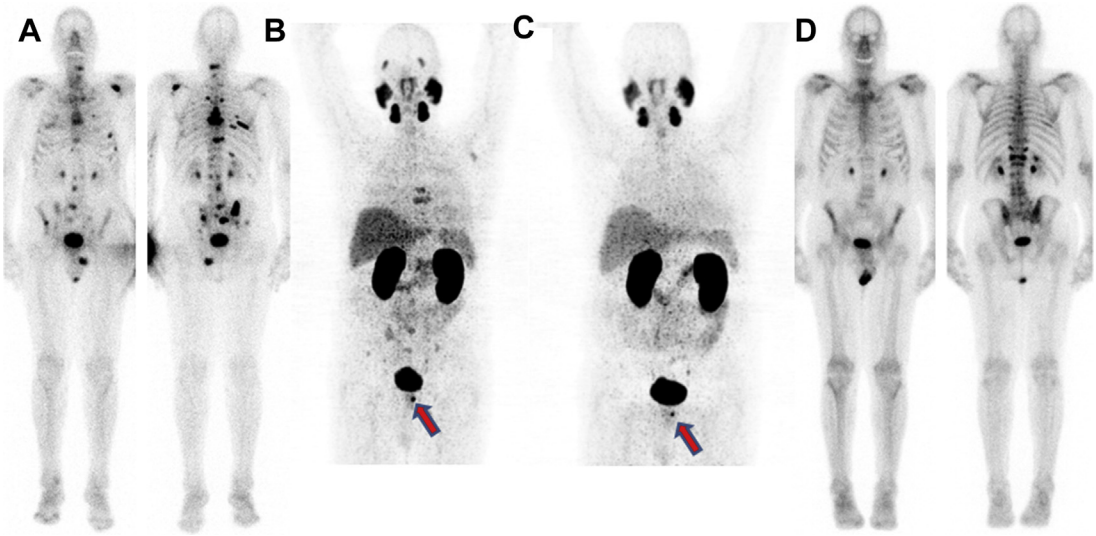

Fig. 1. A 74-year-old patient with CRPC and an increasing PSA level during a running abiraterone and leuprolide acetate therapy. He had a history of radiation therapy for prostate. The bone scintigraphy with ^{99m}Tc-MDP (*A*) showed multiple osteoblastic bone metastases; however, in the same time, the ^{68}Ga-PSMA PET scan showed only a few bone metastases (*B*), which were significantly less in number compared with bone scintigraphy and no lymph node metastases but a small local recurrence (*red arrow*). The PSA and ALP levels prior to the first cycle of ^{223}Ra therapy were 17 ng/mL and 463 U/L, respectively. The patient received 6 cycles of ^{223}Ra and the PSA level decreased continuously during cycles from 17 ng/mL to 1.05 ng/mL (4 weeks after the last cycle). The ALP showed also a decreasing value from 463 to 107 U/L (4 weeks after the last cycle). The PSMA PET (*C*) and bone scintigraphy (*D*) 4 weeks after the last cycle showed a significant response with regression of osteoblastic activity and the PSMA expression.

were measured prior to each treatment cycle and again 4 weeks after the final cycle. In group 1, 4 patients (12.9%) showed a PSA decline, with 1 patient who had a PSA decline greater than 50%. In contrast, in group 2, 14 patients (43.8%) showed a PSA decline, 8 of whom showed a decline greater than 50% (P = .007). Overall, 37 patients had a high ALP level (19 from group 1 and 18 from group 2). Twelve (63.2%) and 16 (88.9%) patients from groups 1 and 2, respectively, showed an ALP decline. This difference was not statistically significant; however, 7 (36%) and 13 (72.2%) patients in groups 1 and 2, respectively, showed an ALP decline greater than 30% (P = .04). The investigators concluded that using ^{68}Ga-PSMA PET as a gatekeeper, radionuclide therapy with ^{223}Ra may be more effective and successful (**Fig. 1**). The determination of a therapy response with a bone scan may take 4 months to 6 months, meaning that patients may stay on potentially ineffective and toxic treatments for longer than necessary.[101] It seems that metabolic imaging, such as ^{68}Ga-PSMA PET, may be a better option compared with a bone scan for the purpose of follow-up.[102,103] This, however, demands prospective studies be performed.

SUMMARY

A pain relief response is seen in approximately one-half of patients treated with radionuclides for painful osseous metastases. The ALSYMPCA study has shown an OS benefit for patients treated with the α-emitter ^{223}Ra. Other radiopharmaceuticals (exclusively β-emitters) discussed in this review have not yet been tested in prospective multicenter trials with large numbers of patients. They have also shown an OS benefit, however, in various studies.

A combination of hormone therapy with bone-targeted therapy may be more effective compared with a single therapy approach. Radionuclide imaging can improve the planning of bone-targeted therapies and, therefore, potentially lead to improvements in OS, PFS, and toxicity. Radionuclide imaging, especially using PET and PET/CT, is one of the most important modalities for the treatment planning and follow-ups.

FDG PET plays an important role in the detection of bony metastases with higher detection rate compared with bone scintigraphy in a majority of tumor types and consistent with clinical status of the patients as it reflects tumor activity. In prostate cancer, PSMA PET together with bone scintigraphy can be used as a gatekeeper for

selection of eligible patients for a radionuclide therapy for bony metastases. Skeletal tumor burden measured by fluoride PET correlates with OS in patients who underwent bone targeted therapies using radionuclides.

REFERENCES

1. Chiang AC, Massague J. Molecular basis of metastasis. N Engl J Med 2008;359:2814–23.
2. Coleman RE. Clinical features of metastatic bone disease and risk of skeletal morbidity. Clin Cancer Res 2006;12:6243s–9s.
3. Yin JJ, Pollock CB, Kelly K. Mechanisms of cancer metastasis to the bone. Cell Res 2005;15: 57–62.
4. Waning DL, Guise TA. Molecular mechanisms of bone metastasis and associated muscle weakness. Clin Cancer Res 2014;20:3071–7.
5. Guise TA, Mohammad KS, Clines G, et al. Basic mechanisms responsible for osteolytic and osteoblastic bone metastases. Clin Cancer Res 2006; 12:6213s–6s.
6. Costa L, Badia X, Chow E, et al. Impact of skeletal complications on patients' quality of life, mobility, and functional independence. Support Care Cancer 2008;16:879–89.
7. Crawford ED, Kozlowski JM, Debruyne FM, et al. The use of strontium 89 for palliation of pain from bone metastases associated with hormone-refractory prostate cancer. Urology 1994;44:481–5.
8. Nair N. Relative efficacy of 32P and 89Sr in palliation in skeletal metastases. J Nucl Med 1999;40: 256–61.
9. Liepe K, Kotzerke J. A comparative study of 188Re-HEDP, 186Re-HEDP, 153Sm-EDTMP and 89Sr in the treatment of painful skeletal metastases. Nucl Med Commun 2007;28:623–30.
10. Yuan J, Liu C, Liu X, et al. Efficacy and safety of 177Lu-EDTMP in bone metastatic pain palliation in breast cancer and hormone refractory prostate cancer: a phase II study. Clin Nucl Med 2013;38: 88–92.
11. Meckel M, Nauth A, Timpe J, et al. Development of a [177Lu]BPAMD labeling kit and an automated synthesis module for routine bone targeted endoradiotherapy. Cancer Biother Radiopharm 2015; 30:94–9.
12. Parker C, Nilsson S, Heinrich D, et al. Alpha emitter radium-223 and survival in metastatic prostate cancer. N Engl J Med 2013;369:213–23.
13. Ahmadzadehfar H. Targeted therapy for metastatic prostate cancer with radionuclides. In: Mohan R, editor. Prostate cancer - leading-edge diagnostic procedures and treatments. Croatia: InTech; 2016.
14. Dietlein M, Eschner W, Grunwald F, et al. Procedure guidelines for radioiodine therapy of differentiated thyroid cancer. Version 4. Nuklearmedizin 2016; 55:77–89 [in German].
15. Franzius C, Schmidt M, Hero B, et al. [Procedure guidelines for MIBG-scintigraphy in children]. Nuklearmedizin 2008;47:132–8.
16. Fendler WP, Kratochwil C, Ahmadzadehfar H, et al. 177Lu-PSMA-617 therapy, dosimetry and follow-up in patients with metastatic castration-resistant prostate cancer. Nuklearmedizin 2016;55:123–8 [in German].
17. Bodei L, Cremonesi M, Grana CM, et al. Peptide receptor radionuclide therapy with (1)(7)(7)Lu-DOTA-TATE: the IEO phase I-II study. Eur J Nucl Med Mol Imaging 2011;38:2125–35.
18. Silberstein EB, Elgazzar AH, Kapilivsky A. Phosphorus-32 radiopharmaceuticals for the treatment of painful osseous metastases. Semin Nucl Med 1992;22:17–27.
19. Lewington VJ, McEwan AJ, Ackery DM, et al. A prospective, randomised double-blind crossover study to examine the efficacy of strontium-89 in pain palliation in patients with advanced prostate cancer metastatic to bone. Eur J Cancer 1991;27: 954–8.
20. Laing AH, Ackery DM, Bayly RJ, et al. Strontium-89 chloride for pain palliation in prostatic skeletal malignancy. Br J Radiol 1991;64:816–22.
21. Mertens WC, Stitt L, Porter AT. Strontium 89 therapy and relief of pain in patients with prostatic carcinoma metastatic to bone: a dose response relationship? Am J Clin Oncol 1993;16:238–42.
22. Mertens WC, Porter AT, Reid RH, et al. Strontium-89 and low-dose infusion cisplatin for patients with hormone refractory prostate carcinoma metastatic to bone: a preliminary report. J Nucl Med 1992; 33:1437–43.
23. Buchali K, Correns HJ, Schuerer M, et al. Results of a double blind study of 89-strontium therapy of skeletal metastases of prostatic carcinoma. Eur J Nucl Med 1988;14:349–51.
24. van der Poel HG, Antonini N, Hoefnagel CA, et al. Serum hemoglobin levels predict response to strontium-89 and rhenium-186-HEDP radionuclide treatment for painful osseous metastases in prostate cancer. Urol Int 2006;77:50–6.
25. Robinson RG, Blake GM, Preston DF, et al. Strontium-89: treatment results and kinetics in patients with painful metastatic prostate and breast cancer in bone. Radiographics 1989;9:271–81.
26. Robinson RG, Preston DF, Schiefelbein M, et al. Strontium 89 therapy for the palliation of pain due to osseous metastases. JAMA 1995;274: 420–4.
27. James ND, Pirrie SJ, Pope AM, et al. Clinical outcomes and survival following treatment of

metastatic castrate-refractory prostate cancer with docetaxel alone or with Strontium-89, Zoledronic Acid, or both: the TRAPEZE randomized clinical trial. JAMA Oncol 2016;2:493–9.
28. James N, Pirrie S, Pope A, et al. TRAPEZE: a randomised controlled trial of the clinical effectiveness and cost-effectiveness of chemotherapy with zoledronic acid, strontium-89, or both, in men with bony metastatic castration-refractory prostate cancer. Health Technol Assess 2016;20:1–288.
29. Windsor PM. Predictors of response to strontium-89 (Metastron) in skeletal metastases from prostate cancer: report of a single centre's 10-year experience. Clin Oncol (R Coll Radiol) 2001;13: 219–27.
30. Oosterhof GO, Roberts JT, de Reijke TM, et al. Strontium(89) chloride versus palliative local field radiotherapy in patients with hormonal escaped prostate cancer: a phase III study of the European Organisation for Research and Treatment of Cancer, Genitourinary Group. Eur Urol 2003;44:519–26.
31. Dafermou A, Colamussi P, Giganti M, et al. A multicentre observational study of radionuclide therapy in patients with painful bone metastases of prostate cancer. Eur J Nucl Med 2001;28: 788–98.
32. Porter AT, McEwan AJ, Powe JE, et al. Results of a randomized phase-III trial to evaluate the efficacy of strontium-89 adjuvant to local field external beam irradiation in the management of endocrine resistant metastatic prostate cancer. Int J Radiat Oncol Biol Phys 1993;25:805–13.
33. Zyskowski A, Lamb D, Morum P, et al. Strontium-89 treatment for prostate cancer bone metastases: does a prostate-specific antigen response predict for improved survival? Australas Radiol 2001;45: 39–42.
34. Lewington VJ. Bone-seeking radionuclides for therapy. J Nucl Med 2005;46(Suppl 1):38S–47S.
35. Maxon HR 3rd, Schroder LE, Hertzberg VS, et al. Rhenium-186(Sn)HEDP for treatment of painful osseous metastases: results of a double-blind crossover comparison with placebo. J Nucl Med 1991;32:1877–81.
36. Liepe K, Franke WG, Kropp J, et al. Comparison of rhenium-188, rhenium-186-HEDP and strontium-89 in palliation of painful bone metastases. Nuklearmedizin 2000;39:146–51 [in German].
37. Sciuto R, Tofani A, Festa A, et al. Short- and long-term effects of 186Re-1,1-hydroxyethylidene diphosphonate in the treatment of painful bone metastases. J Nucl Med 2000;41:647–54.
38. O'Sullivan JM, McCready VR, Flux G, et al. High activity Rhenium-186 HEDP with autologous peripheral blood stem cell rescue: a phase I study in progressive hormone refractory prostate cancer metastatic to bone. Br J Cancer 2002;86:1715–20.
39. Lange R, Overbeek F, de Klerk JM, et al. Treatment of painful bone metastases in prostate and breast cancer patients with the therapeutic radiopharmaceutical rhenium-188-HEDP. Clinical benefit in a real-world study. Nuklearmedizin 2016;55:188–95.
40. Liepe K, Hliscs R, Kropp J, et al. Rhenium-188-HEDP in the palliative treatment of bone metastases. Cancer Biother Radiopharm 2000;15:261–5.
41. Palmedo H, Manka-Waluch A, Albers P, et al. Repeated bone-targeted therapy for hormone-refractory prostate carcinoma: tandomized phase II trial with the new, high-energy radiopharmaceutical rhenium-188 hydroxyethylidenediphosphonate. J Clin Oncol 2003;21:2869–75.
42. Biersack HJ, Palmedo H, Andris A, et al. Palliation and survival after repeated (188)Re-HEDP therapy of hormone-refractory bone metastases of prostate cancer: a retrospective analysis. J Nucl Med 2011; 52:1721–6.
43. Palmedo H, Guhlke S, Bender H, et al. Dose escalation study with rhenium-188 hydroxyethylidene diphosphonate in prostate cancer patients with osseous metastases. Eur J Nucl Med 2000;27: 123–30.
44. van Dodewaard-de Jong JM, de Klerk JMH, Bloemendal HJ, et al. A randomised, phase II study of repeated rhenium-188-HEDP combined with docetaxel and prednisone versus docetaxel and prednisone alone in castration-resistant prostate cancer (CRPC) metastatic to bone; the Taxium II trial. Eur J Nucl Med Mol Imaging 2017;44: 1319–27.
45. Liepe K, Kropp J, Runge R, et al. Therapeutic efficiency of rhenium-188-HEDP in human prostate cancer skeletal metastases. Br J Cancer 2003;89: 625–9.
46. Kolesnikov-Gauthier H, Lemoine N, Tresch-Bruneel E, et al. Efficacy and safety of 153Sm-EDTMP as treatment of painful bone metastasis: a large single-center study. Support Care Cancer 2018;26(3):751–8.
47. Tian JH, Zhang JM, Hou QT, et al. Multicentre trial on the efficacy and toxicity of single-dose samarium-153-ethylene diamine tetramethylene phosphonate as a palliative treatment for painful skeletal metastases in China. Eur J Nucl Med 1999;26:2–7.
48. Sartor O, Reid RH, Hoskin PJ, et al. Samarium-153-Lexidronam complex for treatment of painful bone metastases in hormone-refractory prostate cancer. Urology 2004;63:940–5.
49. Ayati N, Aryana K, Jalilian A, et al. Treatment efficacy of (153)Sm-EDTMP for painful bone metastasis. Asia Ocean J Nucl Med Biol 2013; 1:27–31.
50. Enrique O, Zhongyun P, Parma EP, et al. Efficacy and toxicity of Sm-153 EDTMP in the palliative

treatment of painful bone metastases. World J Nucl Med 2002;1(1):21–7.
51. Borso E, Boni G, Pastina I, et al. Safety and antitumor efficacy of (153)Sm-EDTMP and docetaxel administered sequentially to patients with metastatic castration-resistant prostate cancer. Nucl Med Commun 2014;35:88–94.
52. Autio KA, Pandit-Taskar N, Carrasquillo JA, et al. Repetitively dosed docetaxel and (1)(5)(3) samarium-EDTMP as an antitumor strategy for metastatic castration-resistant prostate cancer. Cancer 2013;119:3186–94.
53. Fizazi K, Beuzeboc P, Lumbroso J, et al. Phase II trial of consolidation docetaxel and samarium-153 in patients with bone metastases from castration-resistant prostate cancer. J Clin Oncol 2009;27:2429–35.
54. Lin J, Sinibaldi VJ, Carducci MA, et al. Phase I trial with a combination of docetaxel and (1)(5)(3)Sm-lexidronam in patients with castration-resistant metastatic prostate cancer. Urol Oncol 2011;29:670–5.
55. Saad F, Carles J, Gillessen S, et al. Radium-223 and concomitant therapies in patients with metastatic castration-resistant prostate cancer: an international, early access, open-label, single-arm phase 3b trial. Lancet Oncol 2016;17:1306–16.
56. Yordanova A, Mayer K, Brossart P, et al. Safety of multiple repeated cycles of 177Lu-octreotate in patients with recurrent neuroendocrine tumour. Eur J Nucl Med Mol Imaging 2017;44:1207–14.
57. Ahmadzadehfar H, Eppard E, Kurpig S, et al. Therapeutic response and side effects of repeated radioligand therapy with 177Lu-PSMA-DKFZ-617 of castrate-resistant metastatic prostate cancer. Oncotarget 2016;7:12477–88.
58. Afshar-Oromieh A, Hetzheim H, Kratochwil C, et al. The theranostic PSMA ligand PSMA-617 in the diagnosis of prostate cancer by PET/CT: biodistribution in humans, radiation dosimetry, and first evaluation of tumor lesions. J Nucl Med 2015;56:1697–705.
59. Fakhari A, Jalilian AR, Yousefnia H, et al. Radiolabeling and evaluation of two 177Lu-labeled bisphosphonates. Iran J Nucl Med 2015;23:108–15.
60. Alavi M, Omidvari S, Mehdizadeh A, et al. Metastatic bone pain palliation using (177)Lu-ethylenediaminetetramethylene phosphonic acid. World J Nucl Med 2015;14:109–15.
61. Agarwal KK, Singla S, Arora G, et al. (177)Lu-EDTMP for palliation of pain from bone metastases in patients with prostate and breast cancer: a phase II study. Eur J Nucl Med Mol Imaging 2015;42:79–88.
62. Dickie GJ, Macfarlane D. Strontium and samarium therapy for bone metastases from prostate carcinoma. Australas Radiol 1999;43:476–9.
63. Baczyk M, Czepczynski R, Milecki P, et al. 89Sr versus 153Sm-EDTMP: comparison of treatment efficacy of painful bone metastases in prostate and breast carcinoma. Nucl Med Commun 2007;28:245–50.
64. Liepe K, Runge R, Kotzerke J. The benefit of bone-seeking radiopharmaceuticals in the treatment of metastatic bone pain. J Cancer Res Clin Oncol 2005;131:60–6.
65. Sharma S, Singh B, Koul A, et al. Comparative therapeutic efficacy of 153Sm-EDTMP and 177Lu-EDTMP for bone pain palliation in patients with skeletal metastases: patients' pain score analysis and personalized dosimetry. Front Med (Lausanne) 2017;4:46.
66. Thapa P, Nikam D, Das T, et al. Clinical efficacy and safety comparison of 177Lu-EDTMP with 153Sm-EDTMP on an equidose basis in patients with painful skeletal metastases. J Nucl Med 2015;56:1513–9.
67. Parker C, Sartor O. Radium-223 in prostate cancer. N Engl J Med 2013;369:1659–60.
68. Hafeez S, Parker C. Radium-223 for the treatment of prostate cancer. Expert Opin Investig Drugs 2013;22:379–87.
69. Anderson RM, Stevens DL, Sumption ND, et al. Effect of linear energy transfer (LET) on the complexity of alpha-particle-induced chromosome aberrations in human CD34+ cells. Radiat Res 2007;167:541–50.
70. Sartor O, Hoskin P, Coleman RE, et al. Chemotherapy following radium-223 dichloride treatment in ALSYMPCA. Prostate 2016;76:905–16.
71. Nilsson S, Cislo P, Sartor O, et al. Patient-reported quality-of-life analysis of radium-223 dichloride from the phase III ALSYMPCA study. Ann Oncol 2016;27:868–74.
72. Nilsson S, Larsen RH, Fossa SD, et al. First clinical experience with alpha-emitting radium-223 in the treatment of skeletal metastases. Clin Cancer Res 2005;11:4451–9.
73. Nilsson S, Franzen L, Parker C, et al. Bone-targeted radium-223 in symptomatic, hormone-refractory prostate cancer: a randomised, multicentre, placebo-controlled phase II study. Lancet Oncol 2007;8:587–94.
74. Parker CC, Pascoe S, Chodacki A, et al. A randomized, double-blind, dose-finding, multicenter, phase 2 study of radium chloride (Ra 223) in patients with bone metastases and castration-resistant prostate cancer. Eur Urol 2013;63:189–97.
75. Pacilio M, Ventroni G, De Vincentis G, et al. Dosimetry of bone metastases in targeted radionuclide therapy with alpha-emitting (223)Ra-dichloride. Eur J Nucl Med Mol Imaging 2016;43:21–33.

76. Sartor O, Heinrich D, Mariados N, et al. Re-treatment with radium-223: first experience from an international, open-label, phase I/II study in patients with castration-resistant prostate cancer and bone metastases. Ann Oncol 2017;28: 2464–71.
77. Etchebehere EC, Milton DR, Araujo JC, et al. Factors affecting (223)Ra therapy: clinical experience after 532 cycles from a single institution. Eur J Nucl Med Mol Imaging 2016;43:8–20.
78. Fosbol MO, Petersen PM, Kjaer A, et al. Radium-223 therapy of advanced metastatic castration-resistant prostate cancer: quantitative assessment of skeletal tumor burden for prognostication of clinical outcome and hematological toxicity. J Nucl Med 2018;59(4):596–602.
79. Rohren EM, Etchebehere EC, Araujo JC, et al. Determination of skeletal tumor burden on 18F-Fluoride PET/CT. J Nucl Med 2015;56: 1507–12.
80. Etchebehere EC, Araujo JC, Fox PS, et al. Prognostic factors in patients treated with 223Ra: the role of skeletal tumor burden on baseline 18F-Fluoride PET/CT in predicting overall survival. J Nucl Med 2015;56:1177–84.
81. Liu FY, Yen TC, Chen MY, et al. Detection of hematogenous bone metastasis in cervical cancer: 18F-fluorodeoxyglucose-positron emission tomography versus computed tomography and magnetic resonance imaging. Cancer 2009;115:5470–80.
82. Mahner S, Schirrmacher S, Brenner W, et al. Comparison between positron emission tomography using 2-[fluorine-18]fluoro-2-deoxy-D-glucose, conventional imaging and computed tomography for staging of breast cancer. Ann Oncol 2008;19: 1249–54.
83. Kim MR, Roh JL, Kim JS, et al. 18F-fluorodeoxyglucose-positron emission tomography and bone scintigraphy for detecting bone metastases in patients with malignancies of the upper aerodigestive tract. Oral Oncol 2008;44:148–52.
84. Costelloe CM, Chuang HH, Madewell JE. FDG PET for the detection of bone metastases: sensitivity, specificity and comparison with other imaging modalities. PET Clin 2010;5:281–95.
85. Qu X, Huang X, Yan W, et al. A meta-analysis of (1)(8)FDG-PET-CT, (1)(8)FDG-PET, MRI and bone scintigraphy for diagnosis of bone metastases in patients with lung cancer. Eur J Radiol 2012;81: 1007–15.
86. Cook GJ, Houston S, Rubens R, et al. Detection of bone metastases in breast cancer by 18FDG PET: differing metabolic activity in osteoblastic and osteolytic lesions. J Clin Oncol 1998;16:3375–9.
87. Israel O, Goldberg A, Nachtigal A, et al. FDG-PET and CT patterns of bone metastases and their relationship to previously administered anti-cancer therapy. Eur J Nucl Med Mol Imaging 2006;33: 1280–4.
88. Samarin A, Hullner M, Queiroz MA, et al. 18F-FDG-PET/MR increases diagnostic confidence in detection of bone metastases compared with 18F-FDG-PET/CT. Nucl Med Commun 2015;36:1165–73.
89. Isasi CR, Moadel RM, Blaufox MD. A meta-analysis of FDG-PET for the evaluation of breast cancer recurrence and metastases. Breast Cancer Res Treat 2005;90:105–12.
90. Shie P, Cardarelli R, Brandon D, et al. Meta-analysis: comparison of F-18 Fluorodeoxyglucose-positron emission tomography and bone scintigraphy in the detection of bone metastases in patients with breast cancer. Clin Nucl Med 2008;33:97–101.
91. Rong J, Wang S, Ding Q, et al. Comparison of 18 FDG PET-CT and bone scintigraphy for detection of bone metastases in breast cancer patients. A meta-analysis. Surg Oncol 2013;22:86–91.
92. Al-Muqbel KM, Yaghan RJ. Effectiveness of 18F-FDG-PET/CT vs bone scintigraphy in treatment response assessment of bone metastases in breast cancer. Medicine (Baltimore) 2016;95: e3753.
93. Morris MJ, Akhurst T, Osman I, et al. Fluorinated deoxyglucose positron emission tomography imaging in progressive metastatic prostate cancer. Urology 2002;59:913–8.
94. Shreve PD, Grossman HB, Gross MD, et al. Metastatic prostate cancer: initial findings of PET with 2-deoxy-2-[F-18]fluoro-D-glucose. Radiology 1996;199:751–6.
95. Yeh SD, Imbriaco M, Larson SM, et al. Detection of bony metastases of androgen-independent prostate cancer by PET-FDG. Nucl Med Biol 1996;23: 693–7.
96. Afshar-Oromieh A, Avtzi E, Giesel FL, et al. The diagnostic value of PET/CT imaging with the (68) Ga-labelled PSMA ligand HBED-CC in the diagnosis of recurrent prostate cancer. Eur J Nucl Med Mol Imaging 2015;42:197–209.
97. Afshar-Oromieh A, Haberkorn U, Eder M, et al. [68Ga]Gallium-labelled PSMA ligand as superior PET tracer for the diagnosis of prostate cancer: comparison with 18F-FECH. Eur J Nucl Med Mol Imaging 2012;39:1085–6.
98. Afshar-Oromieh A, Holland-Letz T, Giesel FL, et al. Diagnostic performance of 68Ga-PSMA-11 (HBED-CC) PET/CT in patients with recurrent prostate cancer: evaluation in 1007 patients. Eur J Nucl Med Mol Imaging 2017;44:1258–68.
99. Bräuer A, Rahbar K, Konnert J, et al. Diagnostic value of additional 68Ga-PSMA-PET before 223Ra-dichloride therapy in patients with metastatic prostate carcinoma. Nuklearmedizin 2017;56: 14–22.

100. Ahmadzadehfar H, Azgomi K, Hauser S, et al. 68Ga-PSMA-11 PET as a gatekeeper for the treatment of metastatic prostate cancer with 223Ra: proof of concept. J Nucl Med 2017;58:438–44.
101. Azad GK, Taylor B, Rubello D, et al. Molecular and functional imaging of bone metastases in breast and prostate cancers: an overview. Clin Nucl Med 2016;41:e44–50.
102. Ahmadzadehfar H, Schlenkhoff CD, Rogenhofer S, et al. 68Ga-PSMA-11 PET represents the tumoricidal effect of 223Ra in a patient with castrate-resistant metastatic prostate cancer. Clin Nucl Med 2016;41:695–6.
103. Hama Y. Antitumor effect of 89Sr for multiple bone metastases of breast cancer: diagnosis by 18F-FDG PET/CT. Clin Nucl Med 2014;39:e290–1.
104. Pandit-Taskar N, Larson SM, Carrasquillo JA. Bone-seeking radiopharmaceuticals for treatment of osseous metastases, Part 1: alpha therapy with 223Ra-dichloride. J Nucl Med 2014;55:268–74.
105. Falkmer U, Jarhult J, Wersall P, et al. A systematic overview of radiation therapy effects in skeletal metastases. Acta Oncol 2003;42:620–33.